Explanation: Why Two Wager Tiers Exist

This page explains the idea behind Safe Bet and Smart Bet. For the full math, the results and the citations, read `docs/white-paper.md`.

Two different questions

A wager decision can answer two different questions. This project treats them as two tools, not one blended answer.

Question 1: "What wager can't hurt me if I'm right?"

This question ignores confidence. It assumes that each opponent makes their most dangerous legal move: they bet everything and answer correctly. It asks which wager still wins outright. This project calls the answer **Tier 1**. The app shows it as **Safe Bet**.

Question 2: "Given how sure I am, what wager gives me the best real chance to win?"

This question needs two more inputs: your confidence in the category, and a model of how opponents wager and answer. This project calls the answer **Tier 2** for Final Jeopardy and the **Daily Double engine** for a mid-game Daily Double. The app shows both as **Smart Bet**.

How ties are scored

Since the 24 November 2014 episode, a tie for first after Final Jeopardy goes to a tiebreaker clue. Only the winner of that clue returns. Smart Bet gives each tied player an equal chance in the tiebreaker. A two-way tie is half a win. A three-way tie is one third of a win. If every player ends at $0 or less, nobody wins.

Tier 1 does not need this rule. It asks for an outright win.

Why Safe Bet and Smart Bet can differ

Safe Bet plans for the worst case every time. Smart Bet uses your confidence and the likely moves of real contestants.

- With low confidence, Smart Bet can recommend less than Safe Bet. A large wager loses more when you miss, and most opponents do not bet the worst case against you.
- A wager that only ties a rival is worth about half a win. One more dollar turns that tie into a win. For this reason, Smart Bet usually takes the extra dollar when a tie is in play.

How Smart Bet models opponents

Smart Bet plays you against **simulated contestants**. The model gives each opponent a role by score: leader, second place, or third place. Each role has a mix of bet types, for example "cover the doubled score of second place" or "bet everything". Some weights come from a published study of real games (Tesauro et al., 2013). Others are our assumptions. `engine/behavior.py` marks each value as PUBLISHED or ASSUMED.

Right and wrong answers are correlated across players. A hard clue is hard for everyone. So when you miss, a rival is more likely to miss too. The model includes you in this correlation.

`docs/adr/0003-simulated-contestant-behavior.md` gives the sources, the decision and its negative consequences.

Why the Daily Double engine gives a range, not one number

A Final Jeopardy wager ends the game. A Daily Double wager changes the rest of the game: more clues, maybe another Daily Double, then Final Jeopardy. No exact formula covers this. The engine simulates the rest of the game thousands of times for each candidate wager. It then counts the win share of each wager.

Each simulated result has some uncertainty, as a poll has a margin of error. When the simulation cannot separate some wagers, the app shows a shaded **plateau** on the chart. The app recommends the smallest wager in the plateau. The evidence does not support a larger risk.

What is proven, and what is assumed

The Safe Bet math is fully determined. It needs no assumption about the accuracy of any player.

Smart Bet and the Daily Double engine need a model of opponents. Part of that model is published: opponent accuracy, the outcome correlation, Daily Double accuracy, and the second-place wager mix in one score regime. The other parts are assumptions: the leader's wager mix, the second-place mix in the other five regimes, the third-place shape, equal tiebreaker odds, and the regular-clue model. The published data are from games played between the mid-1990s and 2012. Treat each Smart Bet percentage as a model output, not as a validated statistic. See `RISK_REGISTER.md` (risks #5, #7 and #14).

Related pages

- `docs/white-paper.md` — the math, the results and the citations.
- `docs/tutorial/getting-started.md` — see both tiers in the app.
- `docs/reference/engine-api.md` — the functions behind these ideas.
- `docs/adr/0003-simulated-contestant-behavior.md` — the opponent model.